

Ilias Test Nr. 3 - Recap

Question 1 (Matching)



Question: Match each statement about activities and technologies to its design implication.

Activities establish requirements for technologies	Start from what must be supported
Contexts establish requirements	Account for where and under which conditions use happens
Technologies offer new ways to undertake activities	Consider opportunities, not only constraints
Design within a sphere of activity	Fit solutions to a domain of practice
Technology changes people's activities	Expect feedback effects after introducing a system

Question 2

Question: During fieldwork, the team photographs forms and follows a document from empty form to final processing. Which two slide principles apply?

Answers:

-  Artifacts provide additional insight into the task.
-  Artifacts should be ignored if interviews were already done.
-  Only final outputs matter
-  All artifacts must be removed from the workplace.
-  Tracing a work product through use can be helpful.

Question 3

Question: For a medical alarm interface, which two activity characteristics are especially relevant?

Answers:

-  Temporal aspects such as timing and frequency.
-  Corporate brand color.
-  Safety criticality: what happens if something goes wrong.
-  Database normalization form.
-  Investor presentation style.

Question 4

Question: The coffee-machine scenario says: put cup under the machine, select options, start preparation, wait, remove full cup. Why is this a conceptual scenario rather than a detailed UI specification?

Answers:

-  It describes user activity without committing to buttons, menus, screens, or sensors.
-  It specifies all device-internal algorithms.
-  It requires a final physical prototype.
-  It contains complete exception handling for all failures.

Question 5

Question: Several user stories about commuters buying different ticket types are combined into a technology-independent description of buying a ticket before travel. What artifact is this most likely to be?

Answers:

-  A conceptual scenario.
-  A high-fidelity prototype.
-  A final physical model.
-  A detailed implementation script.

Question 6

Question: A scenario says: "Maria taps the blue checkout button, scans a QR code, and receives a printed receipt." What type of scenario is this?

Answers:

-  User story, because it only captures anecdotes.
-  Conceptual scenario, because it avoids interface decisions.
-  Concrete scenario, because it suggests particular UI elements and function allocation.
-  Requirement template, because it states project constraints.
-  MoSCoW prioritization, because it ranks features.

Question 7

Question: Match each ticket-machine analysis prompt to the design concern it exposes.

Regular or infrequent?	Frequency of use and learnability burden
Peaks and troughs?	Queueing, throughput, and capacity
Interruptible?	Need for state recovery and clear progress
Safety critical?	Consequences of errors or failures
Mental model to engender	How people should understand the machine and task

Question 8



Question: Match each affordance-related concept with its role in building a mental concept of a system.

Interpreting symbols and components	Necessary to infer how the system can be used
Functionality of the device vs. desired action	Match between what is possible and what the user wants
Semantic distance	Gap between user intention and system meaning
Articulatory distance	Gap between intended action and physical expression
Norman adopting Gibson's concept	Affordance becomes central for perceived use

Question 9



Question: Match each design-analysis purpose with the most suitable type of technique from this chapter.

Estimate how long a pointing action will take	Fitts' Law / physical model
Explain why users misinterpret a system state	Mental models and mapping
Find where a user cannot discover the next possible action	EEAC Gulf of Execution
Represent ordered work steps and alternatives	HTA
Compute expert execution time from low-level actions	KLM

Question 10

Question: A search interface combines query suggestions with clickable result items. Which interpretation best reflects the slide's message?

Answers:

-  Only Power Law of Practice can be applied, because lists are never decisions.
-  The design may involve both decision complexity and pointing effort, so Hick's and Fitts' laws can be considered together.
-  Selection in long lists is purely aesthetic.
-  Hick's law eliminates all pointing costs.

Question 11

Question: A flight-booking form prevents selecting a return date before the departure date. Which design principle is most directly applied?

Answers:

-  A constraint guides the user and minimizes invalid choices.
-  Hick's law is violated because the number of choices increases.
-  The system creates an intentionally inconsistent mapping.
-  KLM predicts that all users will make errors.
-  The affordance is removed because no visible property remains.

Question 12

Question: In the GOMS example for closing a window, the user does not have access to a mouse. Which method should the selection rule predict?

Answer:

-  Use the keyboard shortcut ALT+F4.
-  Use the close button because it is always default even without mouse.
-  Use the context menu because it has the longest method.
-  Perform no action because GOMS cannot model selections.
-  Use Hick's law to remove the window.

Question 13

Question: A GOMS description offers a shortcut method and a menu method for the same goal. Which two concepts explain this representation?

Answers:

-  Methods are sequences of operators or subgoals for accomplishing a goal.
-  Selection rules predict which method is used based on context.
-  Goals are low-level keypresses only.
-  Operators are always complete user intentions rather than actions.
-  GOMS excludes cognitive and perceptual operators.

Question 14

Question: Why does an ATM that returns the card before cash better protect users from forgetting the card?

Answer:

-  It reduces the visual size of the cash slot.
-  It increases decision entropy intentionally.
-  It makes card collection a hidden optional action after goal completion.
-  The overall goal is not yet achieved before the card is collected, so the user remains engaged.
-  It removes all need for feedback.